

Abstract

This seminar paper fills the gap by contributing to the under researched area of AI self-efficacy. It addresses the effects of IT self-efficacy and AI self-efficacy on success attribution and perceived task performance in an academic setting targeting graduate students learning about data science. Our research method is the cohort-level cross-sectional study. The data is analysed by testing four moderated research models using linear regression. In our research findings, the seminar paper identified several limitations including low R^2 values indicating unexplained variance and potential order effects based on the research model design. The conclusions meaningfully contribute to the AI self-efficacy and IT self-efficacy literature and their relationship with success attribution and perceived task performance. AI self-efficacy is found to have a positive relationship with perceived task performance and a comparatively stronger effect compared to IT self-efficacy. The moderating effects of imposter syndrome are studied and found to have an effect on IT self-efficacy.

IT Self-efficacy, AI Self-efficacy, Self-efficacy, Success Attribution, Perceived Task Performance, Attribution Theory, Imposter Syndrome

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Abbreviations

ITSE	Information technology self-efficacy
AISE	Artificial intelligence self-efficacy
SA	Success Attribution
SA_wk1	Success Attribution (week 1)
SA_wk2	Success Attribution (week 2)
PTP	Perceived Task Performance
PTP_wk1	Perceived Task Performance (week 1)
PTP_wk2	Perceived Task Performance (week 2)
AI	Artificial Intelligence
IMP	Imposter Syndrome

1 Introduction

The use of artificial intelligence (AI) is becoming increasingly common in all areas. It is reshaping different aspects of our society, including how we learn and perform. It impacts how students learn new topics and how employees learn new skills for work. It is imperative to study the impact this has on learning behaviour.

This seminar paper observed two cohorts which participated in a study at Friedrich Alexander University, Erlangen-Nuremberg. A group of over 133 graduate students participated in a project-based seminar where they were given two weeks to learn data science techniques using Python, first without and then with assistance of AI tools. This seminar paper uses the data gathered from this project-based seminar and investigates existing research gaps.

This seminar paper builds on social cognitive theory and attribution theory to study the impact of IT self-efficacy and AI self-efficacy on success attribution and perceived task performance. There is existing research studying the effects of an individual's belief in their capabilities on their success attribution and perceived task performance. According to Johnson et al., (2016), IT self-efficacy has an impact on success attribution. Past research also indicates that IT self-efficacy has an impact on perceived task performance (Susanto et al., 2025). However, this study provides unique insight studying the IT self-efficacy effects on success attribution and perceived task performance for graduate students particularly studying data science. Moreover, it also offers unique insights for AI self-efficacy and its potential effects on success attribution and perceived task performance. Additionally, the seminar paper compares the strength of the effects comparing IT self-efficacy's impact to AI self-efficacy's impact. Further insights are provided through a focus on the moderating effects of imposter syndrome. Based on these unique contributions identified from research gaps, this seminar paper aims to answer the following research questions:

Q1. What is the effect of IT Self-Efficacy on Success Attribution and is it moderated by Imposter Syndrome?

Q2: What is the effect of IT Self-Efficacy on Perceived Task Performance and is it moderated by Imposter Syndrome?

Q3. What is the effect of AI Self-Efficacy on Success Attribution and is it moderated by Imposter Syndrome?

Q4: What is the effect of AI Self-Efficacy on Perceived Task Performance and is it moderated by Imposter Syndrome?

Q5: Does AI Self-Efficacy have a stronger effect on Success Attribution than IT Self-Efficacy?

Q6: Does AI Self-Efficacy have a stronger effect on Perceived Task Performance than IT Self-Efficacy?